Anuprita Mhapankar **D20A** **34**

Blockchain Experiment 8

AIM: Implement the Blockchain platform Ganache

THEORY:

Q1: What is Ganache?

Ganache is a personal Ethereum blockchain used for development and testing purposes. It allows developers to deploy smart contracts, test decentralized applications (DApps), and simulate blockchain behavior in a local environment without using real cryptocurrency.

Ganache provides several useful features:

- Instant mining of transactions
- Pre-funded test accounts with Ether
- Detailed transaction logs and block information
- Graphical user interface for monitoring blockchain activity

It is a part of the Truffle Suite and is widely used along with MetaMask and Remix IDE for smart contract development and testing.

Q2: What is a Test Network (Testnet)?

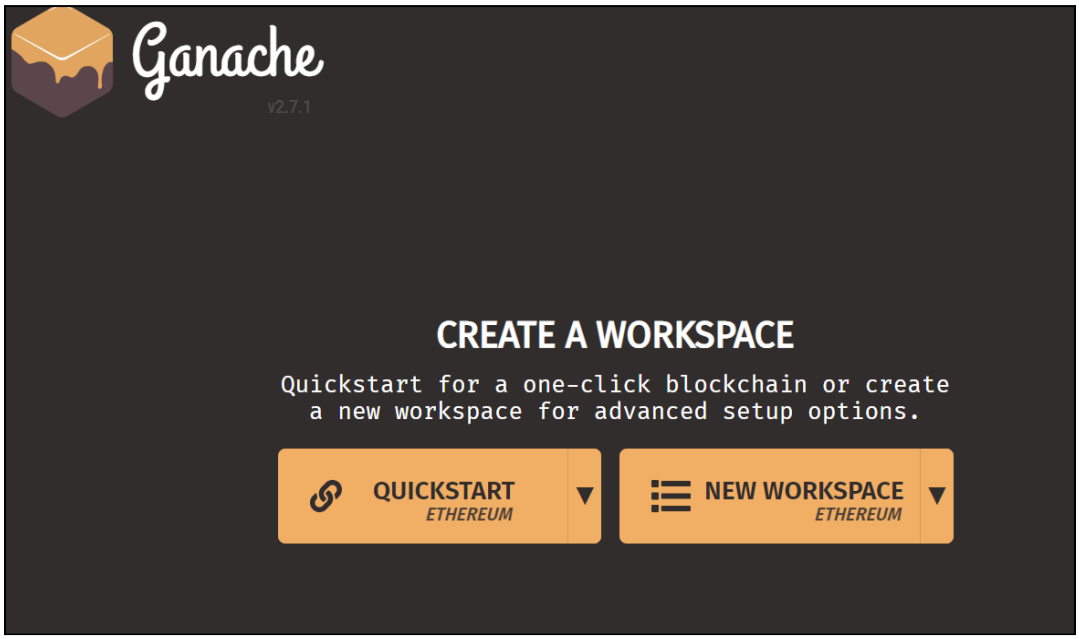
A test network, or testnet, is a blockchain environment used for testing purposes. It simulates the functionality of the main Ethereum network but uses virtual currency instead of real funds. Developers use testnets like Sepolia or Goerli to deploy and test smart contracts without financial risk. Testnets help identify bugs, verify functionality, and ensure proper execution before deploying applications on the mainnet.

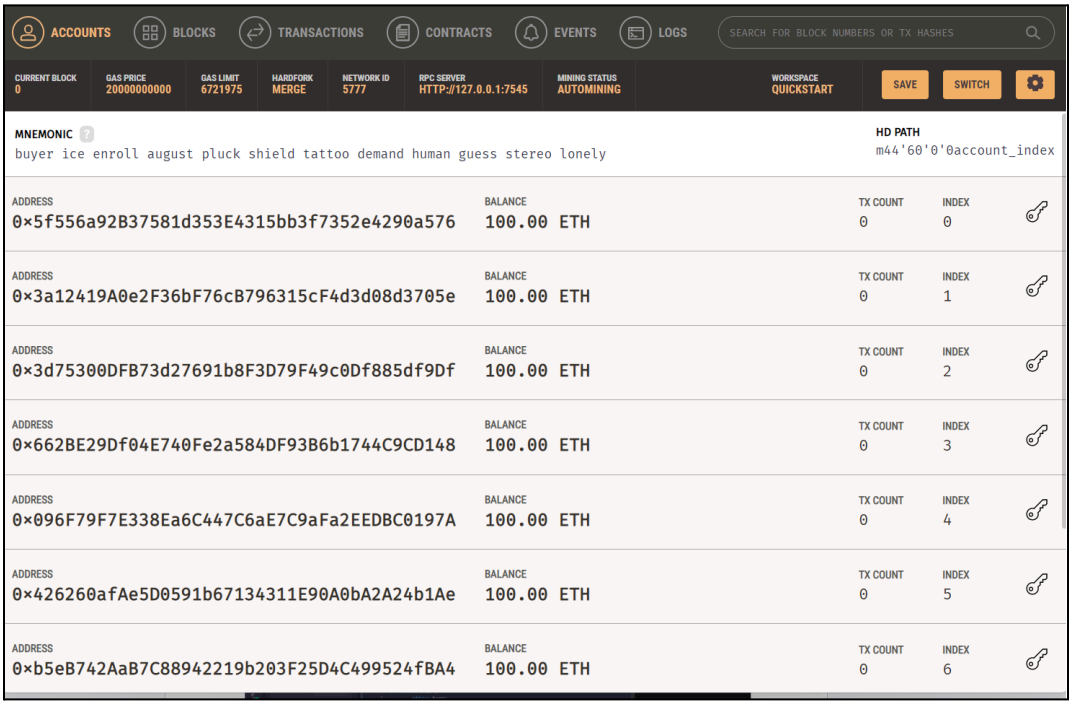
Q3: Steps to Connect MetaMask with Remix IDE for Performing Transactions

- Install the MetaMask extension from the Chrome Web Store
- Create a new wallet or import an existing wallet using a secret recovery phrase
- Enable test networks in MetaMask settings
- Select a test network such as Sepolia
- Add test ETH using a faucet for performing transactions
- Open Remix IDE in the browser (https://remix.ethereum.org)
- Write and compile the smart contract in Remix
- Go to the Deploy & Run Transactions tab
- Select Injected Provider - MetaMask as the environment
- Connect MetaMask account when prompted
- Click Deploy and confirm the transaction in MetaMask
- Interact with deployed contract functions and approve transactions via MetaMask

PROCEDURE

Step 1: Install Ganache and click on Quick Start Ethereum



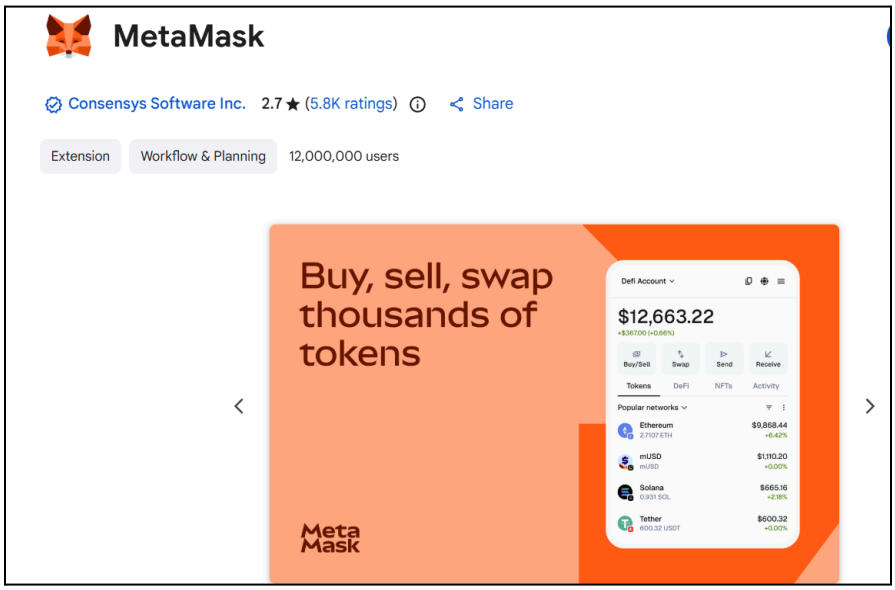


Note the following details from Ganache for later use:

- RPC Server URL: http://127.0.0.1:7545
- Network ID: 5777
- Private keys of accounts (click the key icon next to each address)

Step 2: Install & Setup MetaMask

MetaMask is a browser extension that serves as an Ethereum wallet and allows interaction with decentralized applications. Open Google Chrome and go to the Chrome Web Store.Search for MetaMask (by Consensys Software Inc.) and click Add to Chrome.

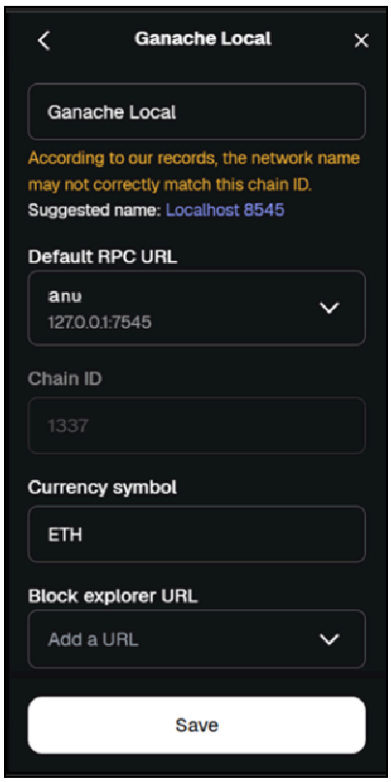


After installation, click 'Create a new wallet' on the MetaMask welcome screen. Set a password and save the Secret Recovery Phrase securely

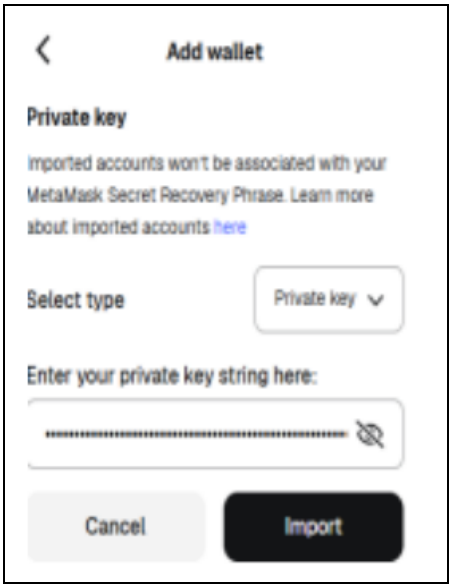
Step 3: Add Ganache Network to MetaMask
In MetaMask, click the network dropdown and select 'Add a custom network'. Enter the following details for the RPC URL:

- RPC URL: http://127.0.0.1:7545
- RPC Name: Ganache Local

Click Save



Step 4: Import Ganache Accounts into MetaMask
In Ganache, click the key icon next to an account address to reveal its private key.
In MetaMask, go to Accounts > Add wallet > Import using Private Key.
Paste the private key and click Import.



Step 5: Create the Smart Contract in Remix IDE

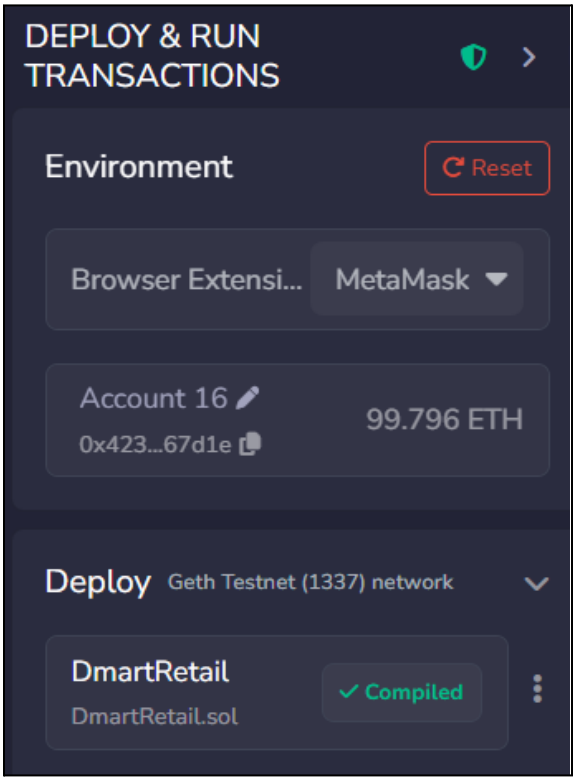
Open https://remix.ethereum.org in the browser.
In the File Explorer, create a new file named DmartRetail.sol. Write the following smart contract:

```
// SPDX-License-Identifier: MIT

pragma solidity ^0.8.0;

contract DmartRetail {

    struct Order {

        address buyer;

        string productIds;

        uint totalAmount;

        uint timestamp;

    }

    struct Review {

        string productId;

        address reviewer;
```

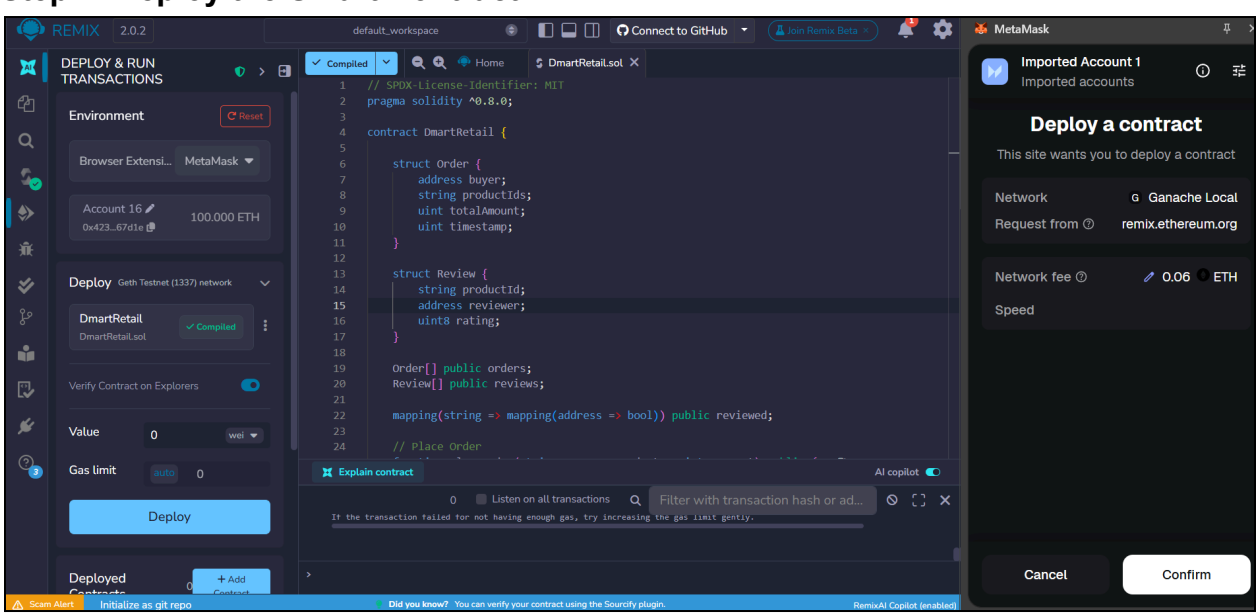
```
        uint8 rating;

    }

    Order[] public orders;

    Review[] public reviews;

    mapping(string => mapping(address => bool)) public reviewed;

    // Place Order

    function placeOrder(string memory _products, uint _amount) public {

        orders.push(Order(msg.sender, _products, _amount,
block.timestamp));

    }

    // Add Review

    function addReview(string memory _productId, uint8 _rating) public {

        require(_rating >= 1 && _rating <= 5, "Invalid rating");

        require(!reviewed[_productId][msg.sender], "Already reviewed");

        reviews.push(Review(_productId, msg.sender, _rating));

        reviewed[_productId][msg.sender] = true;

    }

    // Get total orders

    function getOrderCount() public view returns(uint) {

        return orders.length;

    }

    // Get total reviews

    function getReviewCount() public view returns(uint) {

        return reviews.length;
```

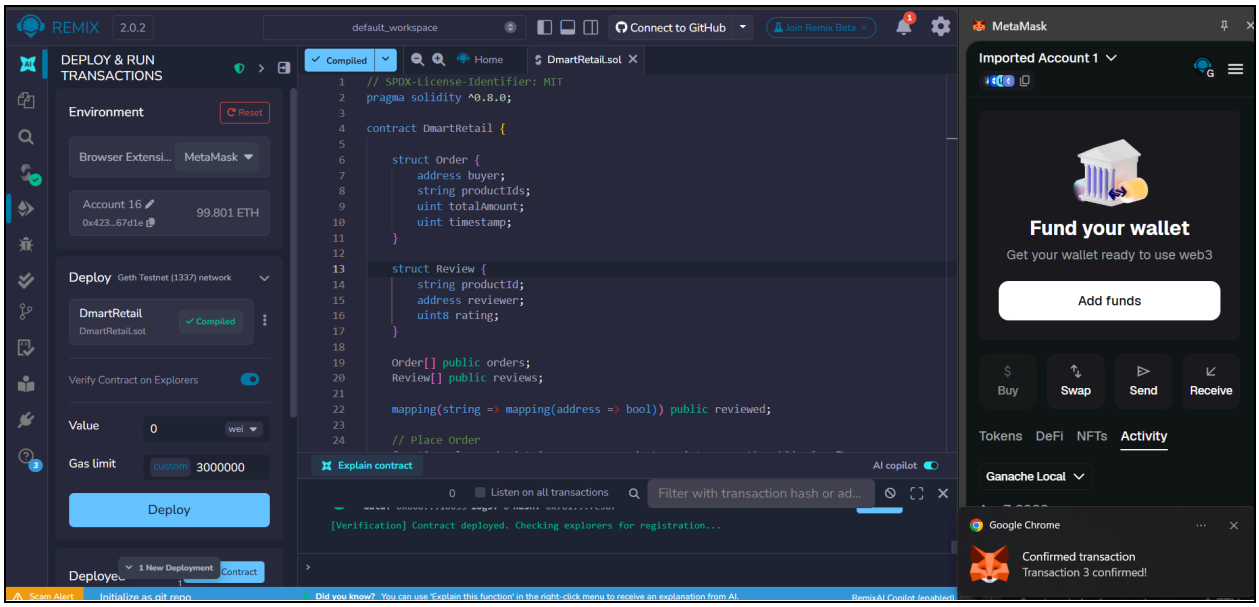
```
    }

}
```

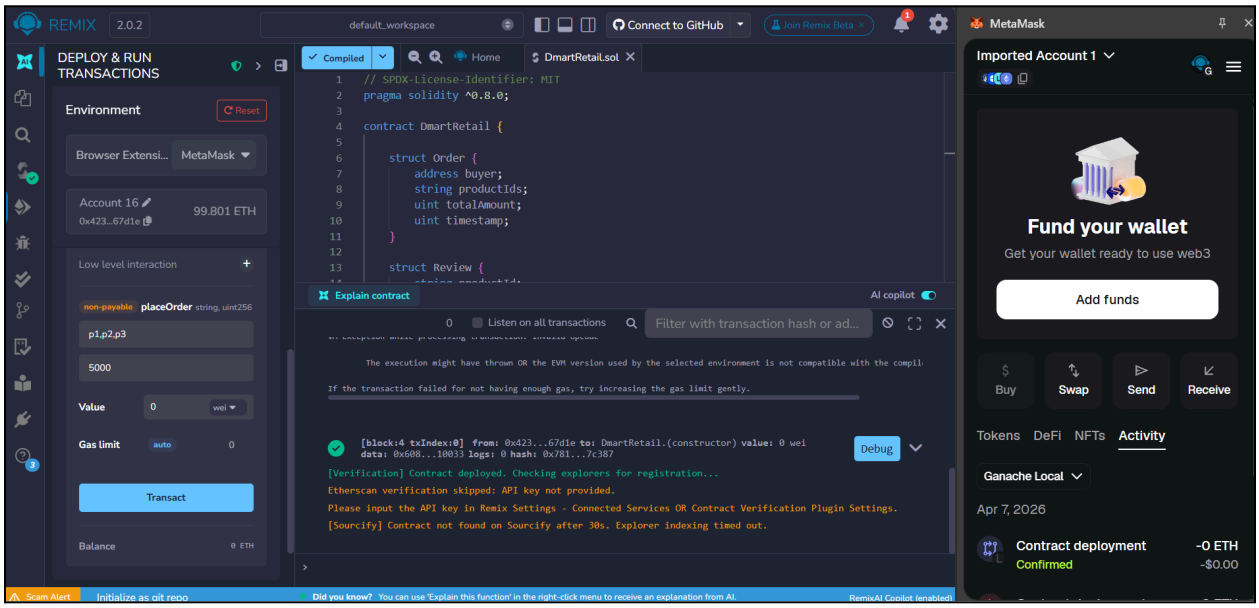
Step 6: Compile the Smart Contract



Step 7: Deploy the Smart Contract

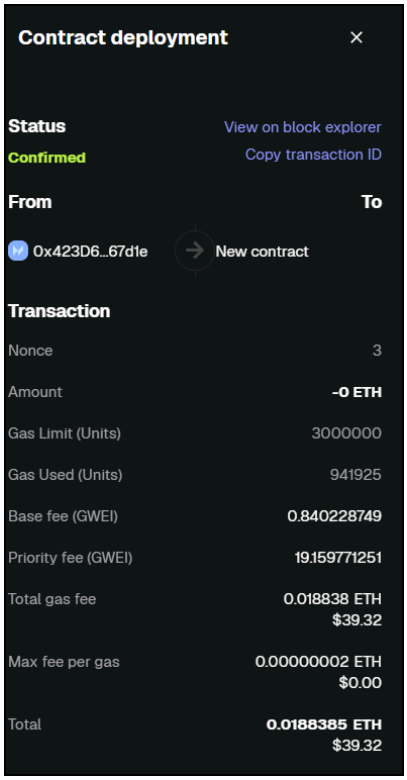






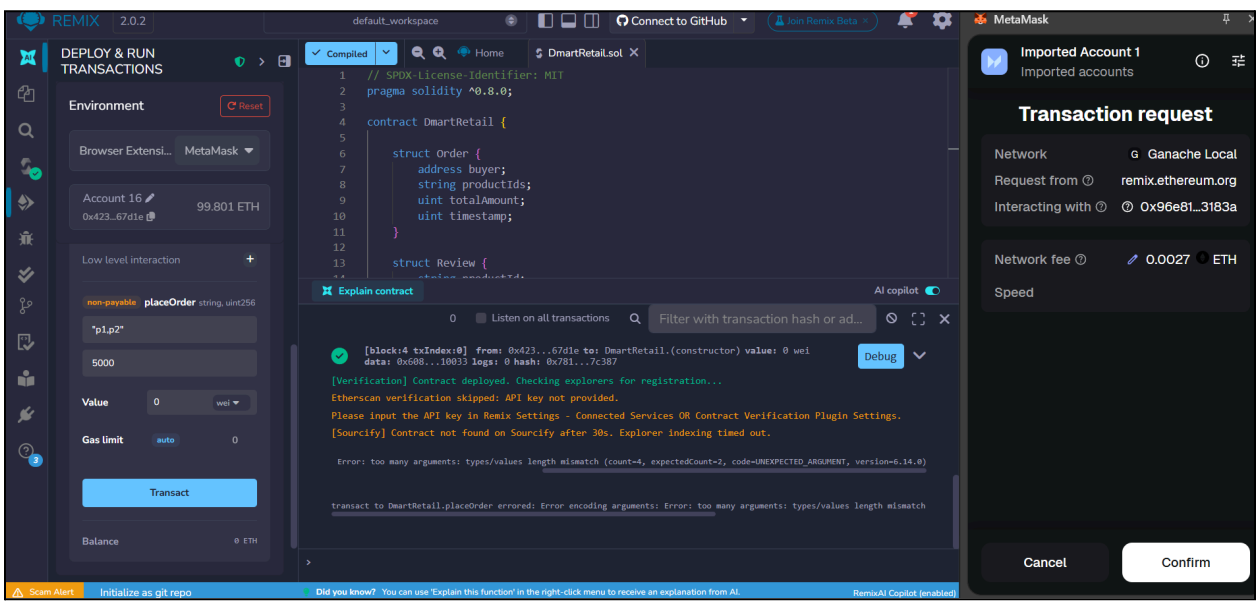
The contract deployment details in MetaMask show:

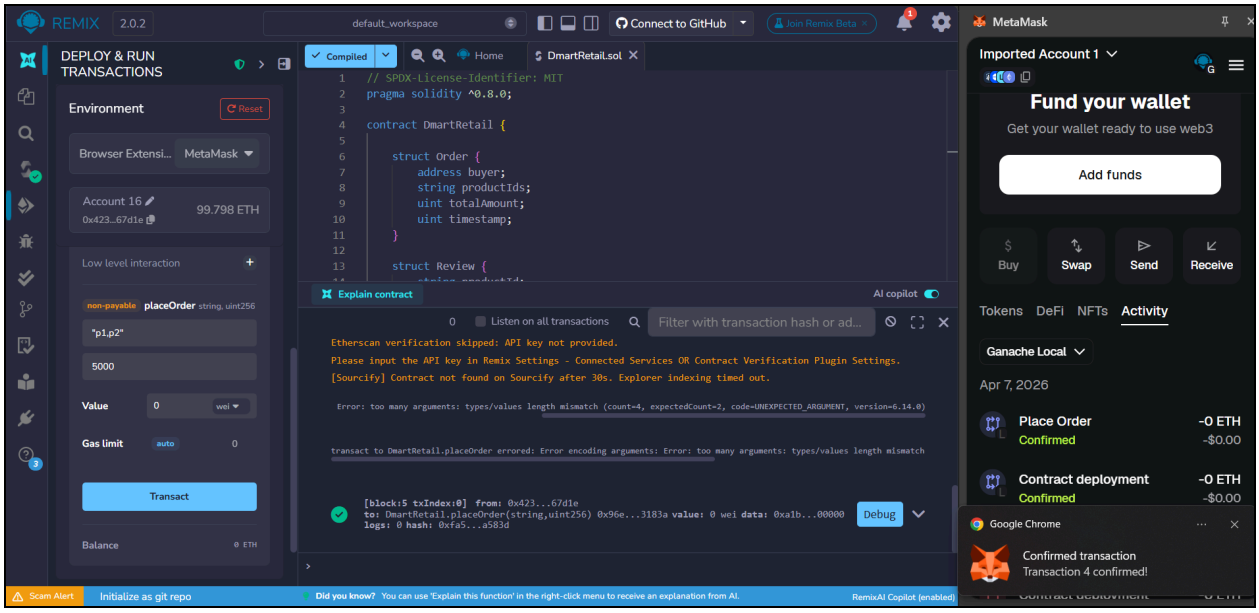
- Status: Confirmed
- From: 0x423D6…67d1e→ To: New contract
- Gas Used: 941,926 units
- Total gas fee: 0.018838 ETH
- Account balance after: 99.80 ETH



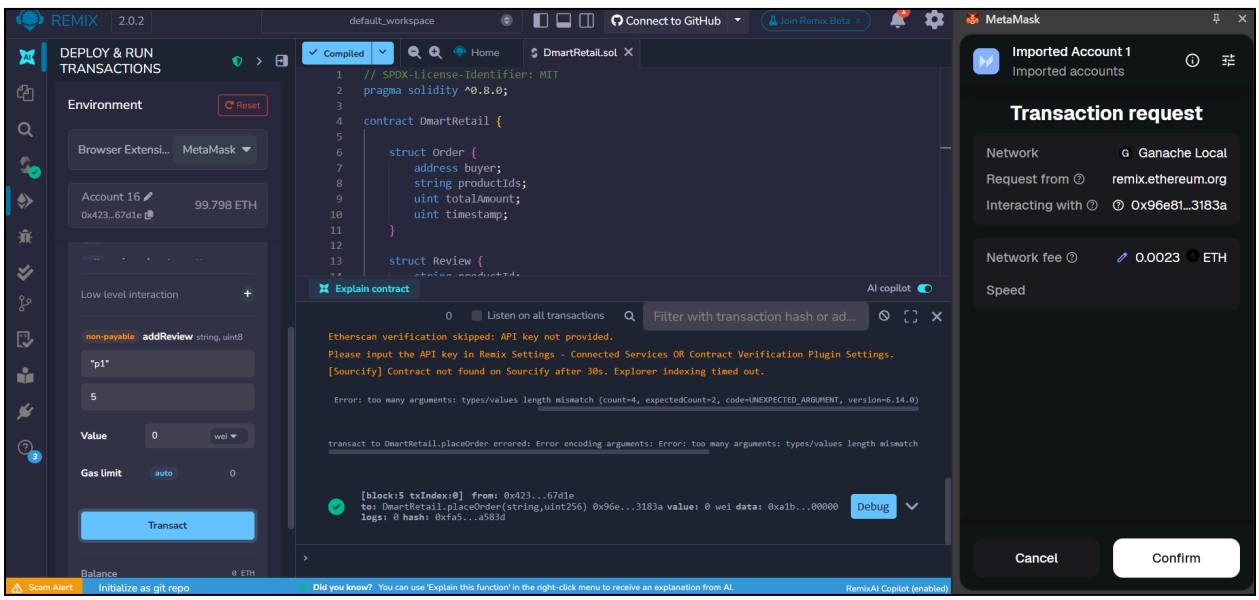
Step 8: Interact with the Smart Contract

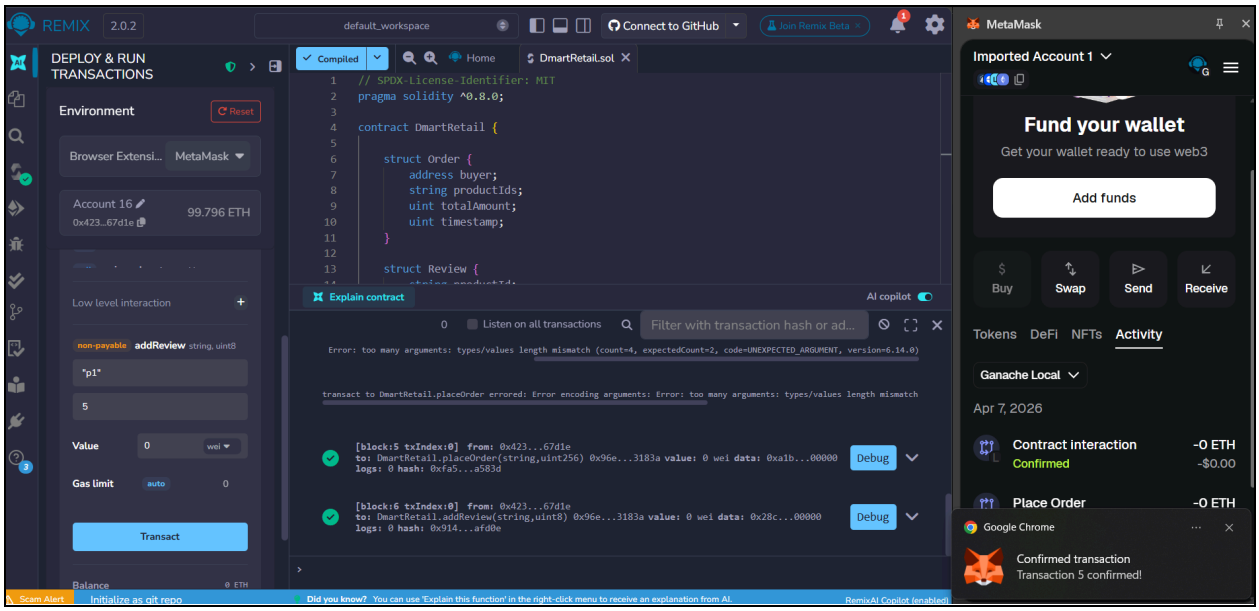
Call placeOrder(p1,p2,5000) where p1,p2 are products and 5000 is the amount.





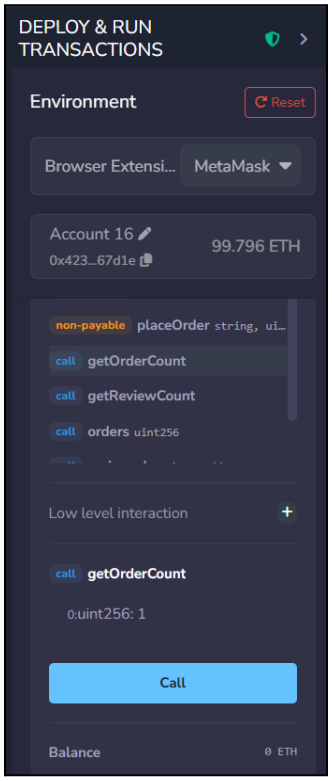
Call addReview where p1 is the product and 5 is the rating(review) given to it.

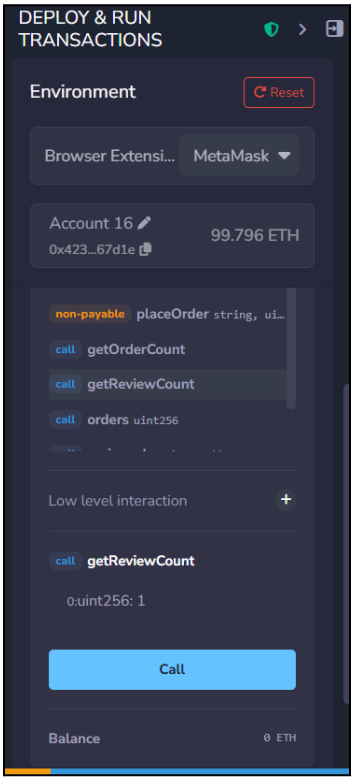




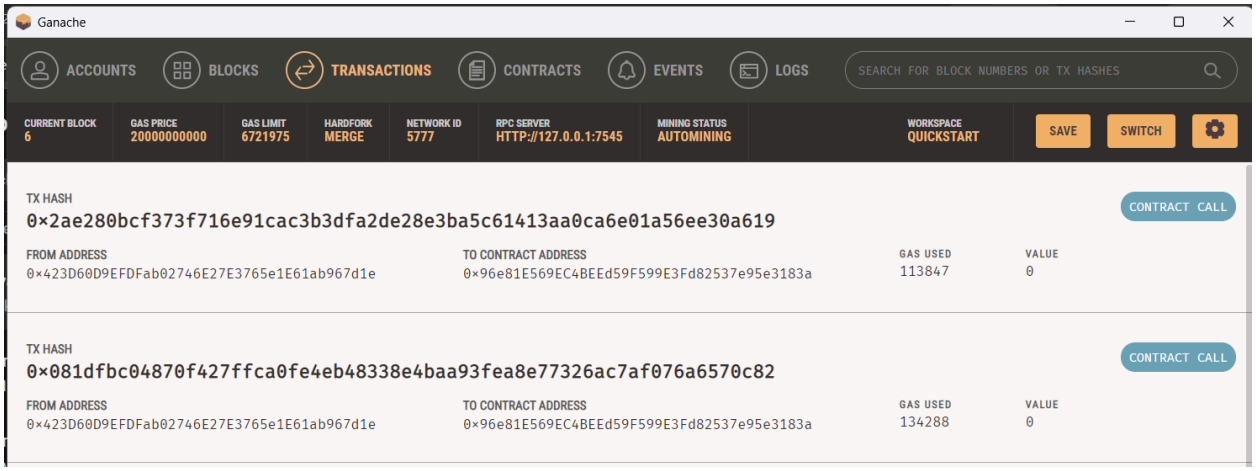
Step 9: Verify Data

Verify data using getOrderCount() and getReviewCount()





Output shown in Ganache:



CONCLUSION:

The experiment was successfully completed using Ganache v2.7.1, MetaMask, and Remix IDE. A pharmaceutical supply chain smart contract (PharmaSupplyChain) was written in Solidity, compiled, and deployed to the local Ganache blockchain via MetaMask. The contract tracked a medicine batch through all four stages of the supply chain — manufacturer, distributor, retailer, and customer — with each stage recorded on-chain. Transaction details were verified in both MetaMask and the Ganache environment, confirming correct gas deduction and block creation.